Tiny Object Detection in Remote Sensing Images Based on Object Reconstruction and Multiple Receptive Field Adaptive Feature Enhancement

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***Abstract*— Tiny object detection in the field of remote sensing has always been a challenging and interesting topic. Despite many researchers working on this problem, it has not been well-solved due to its complexity. In this article, we analyze the reasons for poor performance of deep-learning-based object detection methods for tiny objects in remote sensing images. Moreover, we propose a new remote sensing image tiny object detection network based on object reconstruction and multiple receptive field adaptive feature enhancement module (MRFAFEM), called ORFENet. Detailedly, object reconstruction aims to reduce the information loss of tiny objects within deep neural networks, which is only used in the training phase and can be discarded in the inference phase. MRFAFEM is designed to enhance the features for detecting tiny objects by dynamically adjusting the multiple receptive field features. We have conducted several experiments on the AI-TODv2 and LEVIR-Ship datasets, both of which are proposed for tiny object detection in remote sensing images. The experimental results indicate the effectiveness of the proposed method. Specifically, the proposed ORFENet can achieve an average precision (AP) of 24.8% on the AI-TODv2 dataset and 83.3% AP50 on the LEVIR-Ship dataset. The code will be released at https://github.com/dyl96/ORFENet.**

***Index Terms*— Multiple receptive field adaptive feature enhancement, object reconstruction, remote sensing images, tiny object detection.**

I. Introduction

REMOTE sensing image object detection (RSOD) aims to locate the position of the interest objects from remote sensing images and to identify the class of the objects. RSOD has important applications in many fields, such as military reconnaissance, traffic monitoring, and sea rescue. Therefore, RSOD has become a very hot research issue in the remote sensing field [1], [2], [3], [4], [5], [6].

Traditional object detection methods usually require hand-designed features [7], [8], [9]. However, because of the complexity of the background and the diversity of the objects, traditional object detection methods cannot satisfy the demand due to the limited representation capability [10]. Recently, with the development of deep learning, especially convolutional neural network (CNN), it brings a new opportunity to the task of RSOD. Unlike traditional object detection methods, deep-learning-based object detection methods can automatically learn more robust features. Therefore, such methods tend to have better performance than traditional counterparts.

Many works have introduced deep learning methods into RSOD. Cheng et al. [11] proposed to learn a rotation-invariant CNN model in the R-CNN framework used for multiclass object detection. To achieve more accurate object localization in high-resolution remote sensing images, Long et al. [12] presented an unsupervised score-based bounding box regression method based on the R-CNN framework. Dong et al. [4] introduced a gated context-aware module for object detection, which is used to integrate local context and global context information. Ye et al. [5] adopted adaptive attention fusion mechanism convolutional network for RSOD.

The above methods have achieved good performance in the field of RSOD. However, different from natural images, remote sensing images tend to have a large number of tiny objects with relatively few pixels because of the long imaging distance. As shown in Fig. 1, tiny objects in remote sensing images are much smaller in size compared with normal-sized objects. Despite the important advances in object detection due to the development of deep neural networks, most of the existing object detection methods are designed for normal objects. Tiny object detection in remote sensing images tends to perform less well because their features are not as rich as normal-sized objects. In addition, the downsampling operation in deep neural networks tends to exacerbate the information loss of tiny objects. Moreover, insufficient feature utilization and inappropriate sample assignment in deep-learning-based methods can also affect the performance of tiny object detection. These reasons will bring about more difficulty in localizing and classifying tiny objects. Therefore, the performance of detecting tiny objects in remote sensing images is often unsatisfactory.

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Fig. 1. Examples of tiny objects in remote sensing images from the AI-TODv2 dataset [15]. These objects appear smaller in size compared with the normal-sized objects, with less distinct external features.

To address the issue of tiny object detection, a number of methods have been proposed. Lin et al. [13] improved the performance of tiny object detection from the perspective of multiscale learning. Xu et al. [14] optimized the sample assignment strategy by improving the metric between boxes. M-Center [15] proposed the keypoint-based detector for tiny objects by improving the single center point prior to multiple center points. Xu et al. [16] made some improvements in label assignment so as to solve the problem of poor assignment of positive and negative samples to tiny objects during training. Xu et al. [17] made some improvements on the object prior and label assignment to be more suitable for tiny objects. Ma et al. [18] presented a feature split–merge–enhancement network to emphasize the features of tiny objects in the feature maps and suppress the background information. Chen et al. [19] proposed to enhance tiny object detection using the degraded reconstruction enhancement method. However, there are still two issues affecting tiny object detection that are not well-addressed, which include the information loss of tiny objects within deep networks and the dynamic management of different features from multiple receptive fields.

In this article, our goal is to address the information loss problem of tiny objects and to strengthen the features of different receptive fields required for tiny object detection. Specifically, to mitigate the information loss problem in deep networks, an additional network branch, which is the object reconstruction branch (ORB), is designed. This branch can reconstruct the regions containing tiny objects during training, thereby mitigating the loss of information related to tiny objects. In addition, we design a feature enhancement module that can dynamically adjust the features of different receptive fields. This module has the capacity to capture features from multiple receptive fields according to the input features and dynamically optimize the combination of features from different receptive fields, so as to enhance the features used for tiny object detection.

Our main contributions are summarized as follows.

1) A tiny object detector for remote sensing images is presented, which incorporates an ORB and a multiple receptive field adaptive feature enhancement module (MRFAFEM), to improve the detection performance of tiny objects. Numerous experiments have been conducted on the AI-TODv2 and LEVIR-Ship datasets. The experimental results demonstrated the effectiveness of the proposed method.
2) The ORB is proposed. Using deep networks for tiny object detection often produces the problem of information loss, which leads to poor performance. Therefore, we propose ORB to alleviate this problem. The ORB works on the feature map for detecting tiny objects and mitigates the information loss problem of tiny objects by reconstructing the tiny object regions. Moreover, the proposed ORB is exclusively used during the training phase and can be omitted during the inference phase. Therefore, there is no additional computational cost of ORB.
3) The MRFAFEM is proposed. Different features from different receptive fields are important for tiny object detection, and the importance of features from different receptive fields for tiny object detection is not the same. Therefore, we propose MRFAFEM. Specifically, MRFAFEM can dynamically adjust the features from different receptive fields to enhance the ability to detect tiny objects.

The rest of this article is organized as follows. The related work is presented in Section II. The details of the proposed method are described in Section III. The experimental setup and results are shown in Section IV. The conclusion is given in Section V.

II. Related Work

A. Generic Object Detection

Generic object detectors can be categorized into two groups: single-stage object detectors and two-stage object detectors.

The single-stage object detectors extract features directly from the input image and predict both the location and class of the object. Detailedly, single-stage methods can be further categorized into anchor-based methods and anchor-free methods. Typical single-stage anchor-based object detectors include SSD [20], RetinaNet [25], YOLO series [21], [22], [23], [24], and so on. The fundamental process of such methods involves predefining a set of anchor boxes with various sizes on the feature maps. Then, object detection is achieved by adjusting the anchor boxes that align with the ground-truth object boxes during training, while the object classes are determined through a classification branch. However, in real-world scenarios, the size of objects is often uncertain, making these approaches requiring a series of predefinition anchors less practical. As a result, several anchor-free object detection methods have been developed. Typical anchor-free object detection methods include [26], [27], [28], [29], [30], [31]. CornerNet [26] formulated the object detection problem as the task of identifying a pair of keypoints representing the upper left and lower right corners of an object. It assigns an embedding vector to each keypoint during the prediction process, and matching the embedding vectors associated with the same object enables determination of the final location. In fully convolutional one-stage object detection (FCOS) [27], object detection is formulated as the prediction of both the location of an object and the corresponding category based on the pixel points inside the object. To improve the prediction accuracy, FCOS introduces a cent-ness metric to determine whether a pixel point is positioned at the center of the object. This metric can improve the confidence of the object center pixel. CenterNet [28] formulated object detection as a combination of object center point regression, object box width and height regression, and classification tasks. Specifically, it predicts the offset of a point from the object's center point, the width and height of the object box, and the object's category. Similar to CornerNet, ExtremeNet [29] viewed object detection as a prediction task involving five keypoints: the leftmost, rightmost, topmost, bottommost, and the center point of the object. RepPoints [30] defined the object as a set of points, and the position of the object is obtained by adaptive tuning of the set of points.

In contrast to single-stage object detection methods, two-stage object detection methods initially identify candidate regions within the input image where objects may be present. Subsequently, they classify these candidate regions and regress the bounding box. The process is more complex in comparison to single-stage methods. Typical two-stage object detection methods include RCNN [32], Fast-RCNN [33], Faster-RCNN [34], Cascade-RCNN [35], Mask-RCNN [36], DetectoRS [37], and so on.

B. Tiny Object Detection

The existing methods for tiny object detection can be broadly categorized into the following groups: data augmentation, multiscale learning, feature enhancement, and improved sample assignment strategy.

1) Data Augmentation: This type of method improves the performance of tiny object detection from the perspective of increasing the numbers of tiny objects. To enhance the diversity of the locations of the tiny objects and the match between the detection boxes and the ground-truth boxes, Kisantal et al. [38] proposed to increase the number of tiny objects for training by copying and pasting tiny objects multiple times in the image. Chen et al. [39], on the other hand, proposed an adaptive resampling strategy for data augmentation on the basis of the above methods, and this strategy performs copy and paste considering the context to avoid scale mismatch or background mismatch.

2) Multiscale Learning: Multiscale learning integrates the semantic information from the deep layer with the detailed information from the shallow layer of the deep network. This approach is an effective method to improve the performance of tiny object detection. One of the most prominent multiscale learning methods is feature pyramid network (FPN) [13]. It introduced a network structure that combines deep features with shallow features through both bottom-up and top-down pathways, achieving effective multiscale feature fusion. Subsequently, many multiscale feature fusion methods based on FPN have also been proposed [40], [41], [42], [43], [44]. Moreover, TridenNet [45] designed multiple branch detection heads with different receptive fields to produce feature maps of specific scales.

3) Feature Enhancement: Some methods improve the detection performance of tiny objects through feature enhancement techniques. Haris et al. [46] conducted end-to-end joint training in image super-resolution and object detection. From a different perspective, Li et al. [47] improved the performance of tiny object detection using adversarial learning between a generator and a discriminator to generate high-resolution feature representations for tiny objects. Bai et al. [48] proposed a multitask generative adversarial network for tiny object detection, whose generator is a super-resolution network that can upscale images with blurred details into clearer images with more detailed features. The discriminator contains three tasks: the score prediction task of the super-resolution image is a false image, the object classification task, and the bounding box regression task.

4) Improved Sample Assignment Strategy: A simple but effective method for assigning positive and negative samples in object detection tasks is to determine the intersection over union (IoU) threshold. However, this may result in some low-quality samples, which are even less friendly to tiny objects. Adaptive training sample selection (ATSS) [49] automatically selected the positive and negative samples through the statistical properties of the objects. Xu et al. [14] introduced the dot distance metric to optimize the selection of positive and negative samples. Xu et al. [16] improved the anchor-based tiny object detector by modifying the metric method of ground-truth boxes and anchor boxes during positive and negative sample assignments. The original IoU-based metric is replaced with a Wasserstein distance based on Gaussian distribution, thereby optimizing the quantity and quality of positive samples. Xu et al. [17] modeled the prior boxes or points in a receptive-field-based manner and used the receptive field distance to measure the distance between the ground-truth bounding boxes and the prior to determine the positive and negative samples.

Compared with these methods, the proposed method in this article reduces information loss by reconstructing tiny object regions and improves tiny object detection from the perspective of using multiscale features favorable for tiny object detection. It is worth noting that the ORB does not incur additional inference overhead, and MRFAFEM can dynamically adjust multiscale features.

III. Proposed Method

The overall structure of the proposed method is shown in Fig. 2, including three main parts: the FCOS [27], the ORB, and the MRFAFEM. The proposed method is named ORFENet. In this section, we first review the baseline method

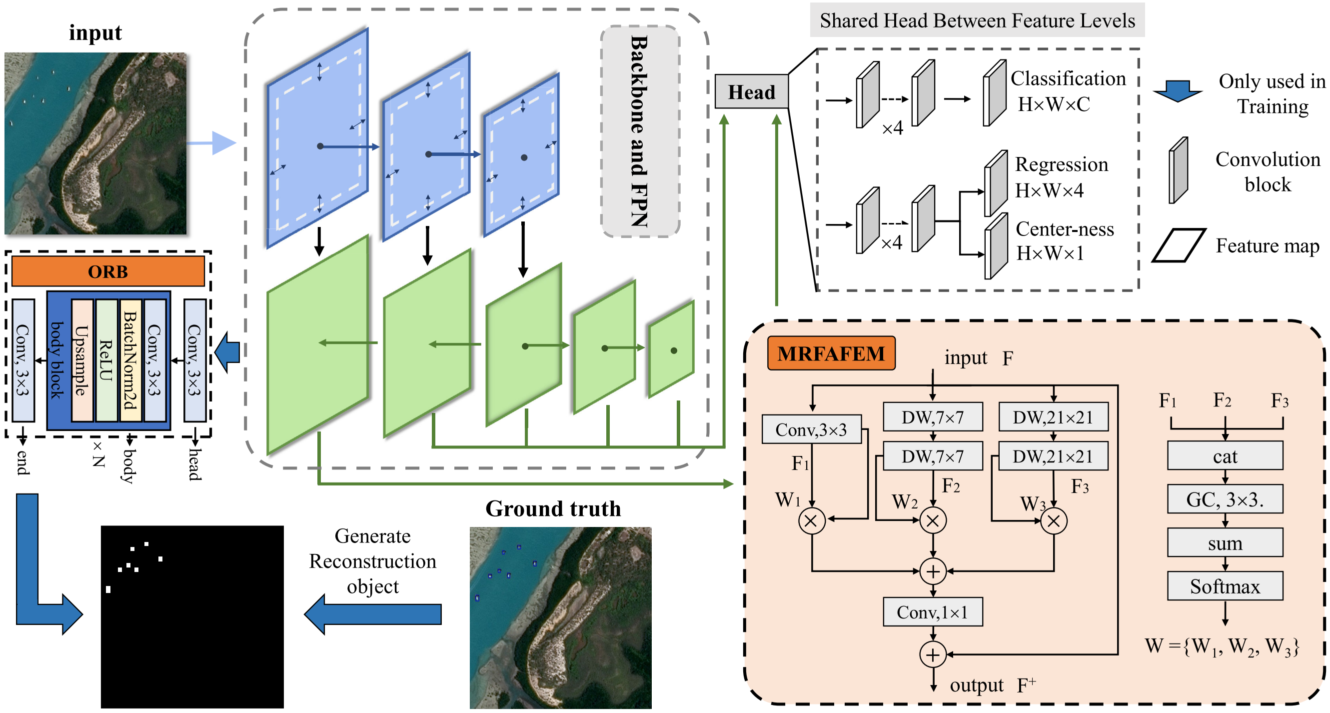


Fig. 2. Overall structure of the proposed ORFENet. ORFENet includes three main parts: the FCOS, the proposed ORB, and the proposed MRFAFEM. The ORB is adopted to suppress the information loss of tiny objects and it is used only in the training phase. The MRFAFAM is used to dynamically enhance the features useful for tiny object detection from different receptive fields. ORB consists of head, body, and end. The head and end are all constricted by the standard 3 × 3 convolutional layer. The value of N is determined based on the size of the input feature map. When the input is the P2 feature map ($H/4$, $W/4$), N is set to 2. When the input feature map is the P3 feature map ($H/8$, $W/8$), N is set to 3. The input of MRFAFEM is F and its output is F^+. The structure contains three branches with different receptive fields. The features of each branch are combined by dynamic weights. DW denotes depthwise convolution. GC denotes group convolution.

FCOS. Then, the proposed ORB and MRFAFEM are explained in detail. Finally, the loss function is described.

A. Revisiting FCOS

FCOS is a typical anchor-free object detection method. FCOS consists of two main parts: the backbone with FPN and the detection head containing classification branch, regression branch, and center-ness branch. When the input image size is $H \times W$, the backbone can obtain a series of features with different sizes $F = \{F_l \in \mathbb{R}^{H_l \times W_l \times C}\}$. Here, l denotes different levels from the backbone and (H_l, W_l) is equal to $(H/2^l, W/2^l)$. These features are then subjected to neighboring feature fusion via FPN to obtain new features $P = \{P_l \in \mathbb{R}^{H_l \times W_l \times C}\}$. In the standard FCOS, l starts from 3. However, for tiny objects, feature maps with higher resolution are especially important due to their more fine-grained features. Therefore, in this article, we not only discuss and study the standard FCOS but also study the FCOS with higher resolution feature maps P_2. The detection heads consist of four 3 × 3 standard convolution layers with a classification head for predicting the class of object, and four 3 × 3 standard convolution layers with regression branch and center-ness branch for localizing the object and estimating the center-ness. Different levels of heads share the same parameters.

B. Object Reconstruction Branch

The proposed ORB in this article serves as an auxiliary task during training, aiming at mitigating the issue of tiny object information loss in its input feature map. During the testing phase, this branch can be safely discarded. This branch operates on the highest resolution feature map output from the FPN. A complex structure for the ORB might lead to a situation where the learning objectives imposed by this branch are primarily focused on the branch itself, rather than effectively constraining the input features. Besides, a complex structure is not well-matched with the comparatively simpler structure of the detection head, which consists of only four convolutional layers. Therefore, we believe that a simple structure is sufficient to meet the requirements for training.

The proposed ORB is implemented with only simple convolutional layers and upsampling operations, and its structure is shown in Fig. 2. The structure of ORB consists of three components: head, body, and end. The head is constricted by a standard 3 × 3 convolutional layer, and the channel number of input features is equal to the channel number of output features. The body comprises N body blocks, each containing convolutional layers, BatchNorm2d, ReLU, and upsampling operations. The upsampling operation is realized by bilinear interpolation. The value of N is determined based on the size of the input feature map. When the input is the P_2 feature map ($H/4$, $W/4$), N is set to 2. When the input feature map is the P_3 feature map ($H/8$, $W/8$), N is set to 3. The end is also implemented by the standard 3 × 3 convolutional layer, but the channels' number of the output feature map is 2. Let the input be X and the output be Y. The ORB can be expressed by the following equation:

$$Y = \text{end}(\text{body}(\text{head}(X))). \tag{1}$$

The reconstruction object is the learning objective of the ORB and also a powerful constraint to suppress the information loss of tiny objects. As shown in Fig. 2, we first generate a mask template with the same size as the input image and all set as 0. Subsequently, based on the positions and sizes of the ground-truth object boxes, we assign values within the mask template to correspond with these ground-truth regions. If a pixel falls within the boundaries of an object box, its value is set to 1; otherwise, it is set to 0. This process enables the generation of the reconstruction object. There are three perspectives that can be used to explain why ORB can work.

1) Suppressing the Information Loss of Tiny Objects: Tiny objects inherently exhibit weaker features, and the series of operations applied to input images by deep networks often result in the loss of feature information related to these tiny objects. This exacerbates the already weak features of these tiny objects. The ORB, by constraining the feature map used for tiny object detection, has the capability to reconstruct the regions corresponding to tiny objects. Consequently, it compels tiny objects to minimize information loss in the feature representation of the input feature map, thereby providing better features for the detection head.

2) Multitask Learning: The ORB and the tiny object detection head can be summarized under the framework of multitask learning. Specifically, the ORB extracts the regions of tiny objects with pixel-level prediction, which has a similar paradigm to some methods for infrared small target detection [50], [51], [52], [53]. The tiny object detection head classifies and regresses the regions of tiny objects in the form of boxes. In the multitask learning framework, two related learning tasks can be mutually reinforcing [54]. Therefore, from that perspective, the proposed ORB can facilitate the learning of the tiny object detection task.

3) Heterogeneous Deep Supervision: Deep supervision is typically introduced by incorporating supervisory information into the middle layers of a network to facilitate the optimization of network parameters. The proposed ORB applies pixel-level supervision to tiny object regions on the FPN feature layers. While this form of supervision differs from the final objectives of bounding box regression and classification, both the learning tasks focus on the regions of tiny objects. Therefore, in a broader sense, the ORB can be considered a form of heterogeneous deep supervision.

C. MRFAFEM

The appearance information of tiny objects is often quite weak, and making more precise judgments based on their surrounding environmental information is advantageous for the detection of tiny objects. For example, the visual characteristics of a tiny ship are extremely limited, and making judgments based solely on its appearance is challenging. Instead, making more discriminative judgments about a tiny ship is facilitated by considering features like its wake during motion, which is more distinctive. In addition, for tiny objects such as storage tanks, making judgments based solely on their individual appearance can be challenging. However, if a tiny storage tank is surrounded by many other storage tanks, this provides a more accurate basis for discrimination. The above content underscores the importance of the surroundings for the precise detection of tiny objects. Moreover, given that the contextual information from different receptive fields is not uniformly applicable for different scenarios involving tiny objects. Some tiny objects can be detected effectively with information from a small receptive field of local region, while others require information from a larger receptive field at a greater distance for accurate judgment. Based on these observations, we designed a dynamic feature enhancement module with multiple receptive fields. This module can dynamically combine features from different receptive fields to enhance the detection of tiny objects.

The detailed structure of MRFFEM is depicted in the lower right corner of Fig. 2. MRFAFEM comprises three branches, each with a different receptive field. These branches are the fine-grained branch with a small receptive field, the close-range context branch with a medium receptive field, and the distant context branch with a large receptive field. The fine-grained branch is implemented with a standard 3×3 convolution, the close-range context branch uses two 7×7 depthwise convolutions, and the distant context branch adopts two 21×21 depthwise convolutions. The computational process of the three branches and the computational process from input to output can be represented by the following equations:

$$F_1 = \text{Conv3}(F) \tag{2}$$
$$F_2 = \text{DW7}(\text{DW7}(F)) \tag{3}$$
$$F_3 = \text{DW21}(\text{DW21}(F)) \tag{4}$$
$$F^+ = F + \text{Conv1}(W_1 F_1 + W_2 F_2 + W_3 F_3) \tag{5}$$

where Conv3 denotes the standard 3×3 convolution, DW7 stands for the 7×7 depthwise convolution, and DW21 represents the 21×21 depthwise convolution. F represents the input feature map, F_1, F_2, and F_3 represent the intermediate output feature maps from the three branches, and F^+ represents the final output feature map. W_1, W_2, and W_3 represent the dynamic weights for the different branches, and their acquisition is illustrated in Fig. 2. Specifically, it can be expressed as

$$\begin{aligned} W &= \text{Softmax}(\text{sum}(\text{GC3}(\text{cat}(F_1, F_2, F_3))) \\ W &= \{W_1, W_2, W_3\} \end{aligned} \tag{6}$$

where cat denotes stacking the feature maps along the channel dimension and GC denotes group convolution with convolution kernel 3×3.

D. Loss Function

The loss function of the proposed method includes two aspects: the object reconstruction loss and the detector loss.

1) Object Reconstruction Loss: As we define the learning objective of the ORB as a classification problem between object foreground and background, we use cross-entropy loss as the loss function. The loss function of the ORB is formulated as follows:

$$\text{Loss}_{\text{or}} = \frac{1}{H \times W} \sum_{i=1}^{H \times W} \text{CE}(\text{pred}, \text{gt}) \tag{7}$$

where gt denotes the mask generated from the ground truth in Fig. 1 and pred denotes the predicted value of the ORB. H and W represent the height and width of the input image, respectively.

2) Detector Loss: The detector loss contains regression loss, classification loss, and center-ness loss. The regression branch is adopted to regress the object bounding box, the classification branch is used to determine the category of the detected object, and the center-ness branch is designed to estimate the importance of each prediction location.

The regression loss Loss_{reg} can be expressed as

$$\text{Loss}_{\text{reg}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} L_{\text{GIOU}}\left(B_i, B_i^*\right) \tag{8}$$

where L_{GIOU} denotes the GIOU loss [59], N_{pos} denotes the number of positive samples, and B_i and B_i^* mean the predicted bounding box and ground-truth bounding box, respectively.

The classification loss is implemented by focal loss. Loss_{cls} can be expressed as

$$\text{Loss}_{\text{cls}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} \text{FL}\left(\boldsymbol{p}_i, \boldsymbol{y}_i^*\right) \tag{9}$$

where N denotes the numbers of total samples, and $\boldsymbol{p}_i$ and $\boldsymbol{y}_i^*$ indicate the predicted probabilities and labels, respectively. FL denotes the focal loss [25].

The center-ness loss $\text{Loss}_{\text{cent}}$ is realized using cross-entropy loss

$$\text{Loss}_{\text{cent}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} \text{CE}\left(\boldsymbol{c}_i, \boldsymbol{c}_i^*\right) \tag{10}$$

where $\boldsymbol{c}_i$ and $\boldsymbol{c}_i^*$ denote the prediction value and ground truth, respectively.

The detector loss Loss_{det} can be summarized as

$$\text{Loss}_{\text{det}} = \text{Loss}_{\text{reg}} + \text{Loss}_{\text{cls}} + \text{Loss}_{\text{cent}}. \tag{11}$$

The total loss of the proposed method is the sum of the object reconstruction loss and the detector loss

$$\text{Loss} = \text{Loss}_{\text{det}} + \lambda \text{Loss}_{\text{or}} \tag{12}$$

where λ is the weight coefficient used to adjust the balance between the object reconstruction loss and the detector loss. In this article, the default value is set to 10.

IV. Experiments

In this section, we will first describe the datasets, evaluation metrics, and the training setting of the experiment. Subsequently, we will provide detailed analysis of the experimental results.

A. Datasets

We conducted several experiments on two public tiny object detection datasets: AI-TODv2 [16] and LEVIR-Ship datasets [19]. The primary experiments are carried out on the challenging AI-TODv2 dataset.

1) AI-TODv2: This dataset serves as a benchmark for evaluating tiny object detection in remote sensing images. It consists of eight different categories, comprising 752 745 object instances across 28 036 remote sensing images. All the images in this dataset are of size 800 × 800 pixels. These images have been collected from various public remote sensing image datasets. What distinguishes this dataset from other remote sensing object detection datasets is that all the objects in this dataset are smaller than 64 × 64 pixels, with 86% of the objects being smaller than 16 × 16 pixels. This dataset includes eight categories: airplane (AI), bridge (BR), storage tank (ST), ship (SH), swimming pool (SP), vehicle (VE), person (PE), and windmill (WM). The number of images in the training set, validation set, and test set is 2/5, 1/10, and 1/2 of the total number of images, respectively. In this article, we merge the training set and validation set together for training, and the test set is used to evaluate the performance.

2) LEVIR-Ship: This dataset contains 3896 images from GF-1/GF-6 and 2320, 788, and 788 images for the training set, validation set, and test set, respectively. The size of the images is all slices of 512 × 512. There are more than 3000 annotated objects of tiny ships in the dataset. The sizes of these objects are primarily below 20 × 20 pixels and are concentrated around 10 × 10 pixels. In this article, we only use the training set for training, while the test set is used to evaluate the performance of different models.

B. Evaluation Metrics

In this article, we evaluate the proposed method using the commonly used average precision (AP), a standard metric in the field of object detection. The AP can be defined as

$$\text{AP} = \int_0^1 P(r)dr \tag{13}$$

where r denotes the recall and $P(r)$ denotes the corresponding precision. The precision and recall are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

where TP, FP, and FN are the true positive, true negative, and false negative, respectively. We use recall as the x-axis and determine the corresponding precision values as the y-axis, which allows us to construct the precision–recall curve. The AP is computed by measuring the area under this precision–recall curve [55]. Specifically, AP50 represents the IoU threshold of 0.5 for defining TP, while AP75 represents the IoU threshold of 0.75 for defining TP. AP denotes the mean value calculated across the range from AP50 to AP95, with IoU intervals set at 0.05.

For the AI-TODv2 dataset, apart from using AP, AP50, and AP75 for evaluation, we also use AP_{vt}, AP_{t}, AP_{s}, and AP_{m} to

TABLE I

EXPERIMENTAL RESULTS OF THE PROPOSED ORB AND MREAFEM, USING FCOS AND FCOS(P2) AS BASELINES, ON THE AI-TODv2 DATASET. ALL EXPERIMENTS WERE TRAINED FOR 12 EPOCHS. PARAMS DENOTES THE PARAMETERS AND FLOPs DENOTE THE FLOATING-POINT OPERATIONS

Baseline	ORB	MRFAFEM	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_t(%)	AP_s(%)	AP_m(%)	Params(M)	FLOPs(G)
FCOS	×	×	13.2	32.1	8.4	2.9	12.4	18.1	27.2	31.85	123.16
	√	×	14.6	36.4	9.1	2.6	13	20.3	**31.2**	31.85	123.16
	×	√	13.9	33.6	9.0	2.6	12.5	19.4	28.3	32.7	131.69
	√	√	**15.1**	**37.6**	**9.8**	**3.1**	**13.4**	**21.3**	31.1	32.7	131.69
FCOS(P2)	×	×	17.3	39.9	12.2	5.6	17.7	21.2	25.4	31.92	339.24
	√	×	18.5	44.1	12.3	5.9	**18.8**	22.6	28.4	31.92	339.24
	×	√	18.4	44.3	12.1	6.7	18.6	22.9	28.3	32.77	373.34
	√	√	**18.9**	**44.4**	**12.7**	**6.9**	18.4	**23.4**	**30.3**	32.77	373.34

evaluate the performance of very tiny, tiny, small, and medium objects, respectively. These metrics can be referenced in [15].

For the experimental results on the LEVIR-Ship dataset, AP50 is exclusively used for evaluation [19].

C. Training Setting

All our experiments have been conducted on a single NVIDIA RTX A6000 GPU. The model code is built on the basis of MMDetection [57], an object detection library based on PyTorch frame [56]. We use the ImageNet [58] pretrained ResNet-50 with FPN as the backbone unless specified otherwise. All the models are trained using the stochastic gradient descent (SGD) optimizer with 0.9 momenta, 0.0001 weight decay, and two batch sizes. The initial learning rate is set to 0.005. The constant warm-up strategy with a ratio of 0.33 is adopted at the first 10 000 iterations.

D. Experimental Results

In this section, we will demonstrate the benefits of our proposed method for tiny object detection. Several sets of experiments are reported, including improvements over the baseline, ablation study, comparisons with other methods, and the results on the LEVIR-Ship dataset.

1) Improvement Over Baseline: To demonstrate the effectiveness of the proposed method, we explore the effectiveness of ORB and MRFAFEM on the AI-TODv2 dataset using FCOS and FCOS(P2) as baseline, respectively. FCOS(P2) indicates that P2 feature maps were used in FPN. The ResNet-50 pretrained on ImageNet is adopted as the backbone. The experimental results are shown in Table I.

From the table, it can be seen that when the proposed ORB is incorporated into the FCOS, the AP is improved by 1.4% compared with the original FCOS, and when the proposed MRFAFEM is incorporated into the FCOS, the AP is improved by 0.7% compared with the original FCOS. When both the proposed ORB and MRFAFEM are simultaneously added to the FCOS, all the metrics show improvement. Specifically, compared with the original FCOS, AP, AP50, and AP75 are improved by 1.9%, 5.5%, and 1.4%, respectively. Similarly, when the proposed ORB is incorporated into FCOS(P2), the AP is improved by 1.2% compared with the original FCOS(P2), and when the proposed MRFAFEM is incorporated into the FCOS(P2), the AP is improved by 1.1% compared with the original FCOS(P2). When both the proposed ORB and MRFAFEM are simultaneously added to the FCOS(P2) algorithm, AP, AP50, and AP75 are improved by 1.6%, 4.5%, and 0.5%, respectively compared with the original FCOS(P2). The data above indicate that the proposed methods lead to improved accuracy compared with the baseline, demonstrating the effectiveness of the proposed approach. Moreover, the incorporation of the P2 feature map also has a certain effect

TABLE II

ABLATION EXPERIMENTS ON THE KERNEL SIZES OF CLOSE-RANGE CONTEXT BRANCH AND DISTANT CONTEXT BRANCH IN MRFAFEM. THE EXPERIMENTS WERE CONDUCTED ON THE AI-TODv2 DATASET. CCB INDICATES THE KERNEL SIZE OF CLOSE-RANGE CONTEXT BRANCH AND DCB DENOTES THE KERNEL SIZE OF DISTANT CONTEXT BRANCH

Method	CCB	DCB	AP(%)	AP50(%)	AP75(%)
FCOS(P2)	×	×	17.3	39.9	12.2
FCOS(P2) w/ MRFAFEM	5	21	18.1	42.3	**12.4**
	7	21	**18.5**	**44.1**	12.3
	9	21	18.2	42.8	12.3
	7	17	17.6	42.6	11.6
	7	21	**18.5**	**44.1**	**12.3**
	7	25	18.3	44.3	11.8

TABLE III

ABLATION EXPERIMENTS ON THE EFFECT OF DYNAMIC WEIGHTS ON PERFORMANCE IN MRFAFEM. THE EXPERIMENTS WERE CONDUCTED ON THE AI-TODv2 DATASET. MRF INDICATES THE MRFAFEM WITHOUT THE DYNAMIC WEIGHT

Baseline	MRF	Weight	AP(%)	AP50(%)	AP75(%)
FCOS(P2)	×	×	17.3	39.9	**12.2**
	√	×	18.2	43.3	12.0
	√	√	**18.4**	**44.3**	12.1

TABLE IV

ABLATION STUDY ON THE IMPACT OF DIFFERENT λ VALUES IN THE LOSS FUNCTION ON THE FINAL PERFORMANCE OF ORFENET. THE EXPERIMENTS WERE CONDUCTED ON THE AI-TODv2 DATASET. $\lambda = 0$ INDICATES THAT THE ORB WAS NOT INCLUDED

Baseline	λ	AP(%)	AP50(%)	AP75(%)
FCOS(P2)	0	17.3	39.9	12.2
	1	17.4	41.4	11.6
	5	17.9	42.1	12.0
	10	**18.5**	**44.1**	12.3
	20	18.4	43.4	**12.5**

TABLE V

ALL-CLASS PERFORMANCE OF THE PROPOSED ORFENET AND OTHER STATE-OF-THE-ART METHODS ON THE AI-TOD-V2 TEST SET. BOLD FONTS INDICATE THE BEST PERFORMANCE. IN THE LAST ROW, *DENOTES THE PROPOSED METHOD WAS TRAINED FOR 36 EPOCHS. THE PROPOSED ORFENET* ACHIEVES THE STATE-OF-THE-ART PERFORMANCE OF 24.8% AP

Method	Backbone	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_t(%)	AP_s(%)	AP_m(%)
Two-stage:								
TridentNet [45]	ResNet-50	10.1	24.5	6.7	0.1	6.3	19.8	31.9
Faster R-CNN w/ ATSS [34][49]	ResNet-50-FPN	12.8	29.6	9.2	0.0	9.2	24.9	36.6
Faster R-CNN [34]	ResNet-50-FPN	12.8	29.9	9.4	0.0	9.2	24.6	37.0
Faster R-CNN [34]	ResNet-101-FPN	13.1	30.7	9.2	0.0	9.7	24.6	35.5
Faster R-CNN [34]	HRNet-w32	14.5	32.8	10.6	0.1	11.1	27.4	37.8
Cascade-R-CNN [35]	ResNet-50-FPN	15.1	34.2	11.2	0.1	11.5	26.7	38.5
DetectoRS [37]	ResNet-50-FPN	16.1	35.5	12.5	0.1	12.6	28.3	**40.0**
DotD [14]	ResNet-50-FPN	20.4	51.4	12.3	8.5	21.1	24.6	30.4
Cascade-R-CNN w/ NWD-RKA [16]	ResNet-50-FPN	22.2	52.5	15.1	7.8	21.8	28.0	37.2
One-stage:								
FCOS [27]	ResNet-50-FPN	13.2	32.1	8.4	2.9	12.4	18.1	27.2
ATSS [49]	ResNet-50-FPN	13.0	31.0	8.7	2.3	11.2	18.0	29.9
RetinaNet [25]	ResNet-50-FPN	8.9	24.2	4.6	2.7	8.4	13.1	20.4
RepPoints [30]	ResNet-50-FPN	9.3	23.6	5.4	2.8	10.0	12.3	18.9
FoveaBox [31]	ResNet-50-FPN	11.3	28.1	7.4	1.4	8.6	17.8	32.2
Grid R-CNN [60]	ResNet-50-FPN	14.3	31.1	11.0	0.1	11.0	25.7	36.7
FSANet [61]	ResNet-50-FPN	17.6	45.0	10.5	5.4	15.8	22.9	33.8
Ours:								
ORFENet	ResNet-50-FPN	18.9	44.4	12.7	6.9	18.4	23.4	30.3
ORFENet*	ResNet-50-FPN	**24.8**	**55.4**	**18.2**	**9.7**	**24.4**	**28.7**	35.1

TABLE VI

CLASSWISE AP OF THE PROPOSED ORFENET AND OTHER EXCELLENT METHODS ON THE AI-TOD-V2 TEST SET. BOLD FONTS INDICATE THE BEST PERFORMANCE. IN THIS TABLE, AP IS THE EVALUATION METRIC OF EACH CLASS. THE SHORT NAMES FOR CATEGORIES ARE DEFINED AS: AIRPLANE (AI), BRIDGE (BR), STORAGE TANK (ST), SHIP (SH), SWIMMING POOL (SP), VEHICLE (VE), PERSON (PE), AND WINDMILL (WM). SPECIFICALLY, THE PROPOSED ORFENET CAN ACHIEVE THE BEST PERFORMANCE IN THE CLASSES OF BR, SH, SP, VE, PE, AND WM

Method	Backbone	AI	BR	ST	SH	SP	VE	PE	WM
Two-stage:									
TridentNet [45]	ResNet-50	19.3	0.1	17.2	16.2	12.4	12.5	3.4	0.0
Faster R-CNN w/ ATSS [34][49]	ResNet-50-FPN	24.7	5.5	20.5	19.8	12.0	15.1	4.8	0.0
Faster R-CNN [34]	ResNet-50-FPN	19.7	4.8	19.0	19.9	3.7	14.4	4.8	0.0
Faster R-CNN [34]	ResNet-101-FPN	25.3	8.5	19.4	19.9	12.5	14.6	4.5	0.0
Faster R-CNN [34]	HRNet-w32	27.9	9.5	21.5	21.4	13.0	16.7	5.9	0.0
Cascade-R-CNN [35]	ResNet-50-FPN	26.2	9.6	24.0	24.3	13.2	17.5	5.8	0.1
DetectoRS [37]	ResNet-50-FPN	**28.5**	11.7	23.2	26.4	14.9	17.6	6.5	0.2
DotD [14]	ResNet-50-FPN	18.7	17.5	34.7	37.0	12.4	25.4	10.3	7.4
Cascade-R-CNN w/ NWD-RKA [16]	ResNet-50-FPN	28.5	17.5	**36.9**	38.3	13.7	26.6	10.4	5.7
One-stage:									
FCOS [27]	ResNet-50-FPN	11.1	13.6	20.5	30.0	7.3	18.0	4.8	0.0
ATSS [49]	ResNet-50-FPN	15.4	11.7	20.0	27.6	9.4	14.8	4.7	0.0
RetinaNet [25]	ResNet-50-FPN	1.3	11.8	14.3	23.6	5.8	11.4	2.3	0.5
RepPoints [30]	ResNet-50-FPN	0.0	0.1	22.5	28.8	0.2	18.3	4.1	0.0
FoveaBox [31]	ResNet-50-FPN	15.6	3.3	21.1	20.8	9.7	16.3	4.0	0.0
Grid R-CNN [60]	ResNet-50-FPN	24.5	11.7	20.9	23.5	12.1	16.1	5.1	0.4
FSANet [61]	ResNet-50-FPN	19.2	16.0	28.3	33.0	12.9	20.4	6.0	5.3
Ours:									
ORFENet	ResNet-50-FPN	14.6	18.8	32.2	38.2	13.1	25.5	8.4	0.0
ORFENet*	ResNet-50-FPN	26.0	**21.1**	35.8	**50.6**	**17.1**	**27.8**	**11.0**	**8.7**

on improving the accuracy of tiny object detection. Therefore, in the subsequent experiments, the FCOS(P2) method with both the proposed ORB and MRFAFEM is referred to as ORFENet.

Furthermore, in terms of algorithm complexity, when the proposed ORB is applied to FCOS, there is no change in Params and floating-point operations (FLOPs), yet there is a significant performance improvement. When the proposed MRFAFEM is applied to FCOS, there is only a minor change in Params and FLOPs, and there is a performance improvement. When ORB and MRFAFEM are used simultaneously, the change in Params and FLOPs is the same as when only MRFAFEM is added, and the performance reaches the peak. Therefore, it can be concluded that ORB can enhance the performance of tiny object detection without introducing additional computational complexity, while MRFAFEM can improve performance with minor changes in Params and FLOPs.

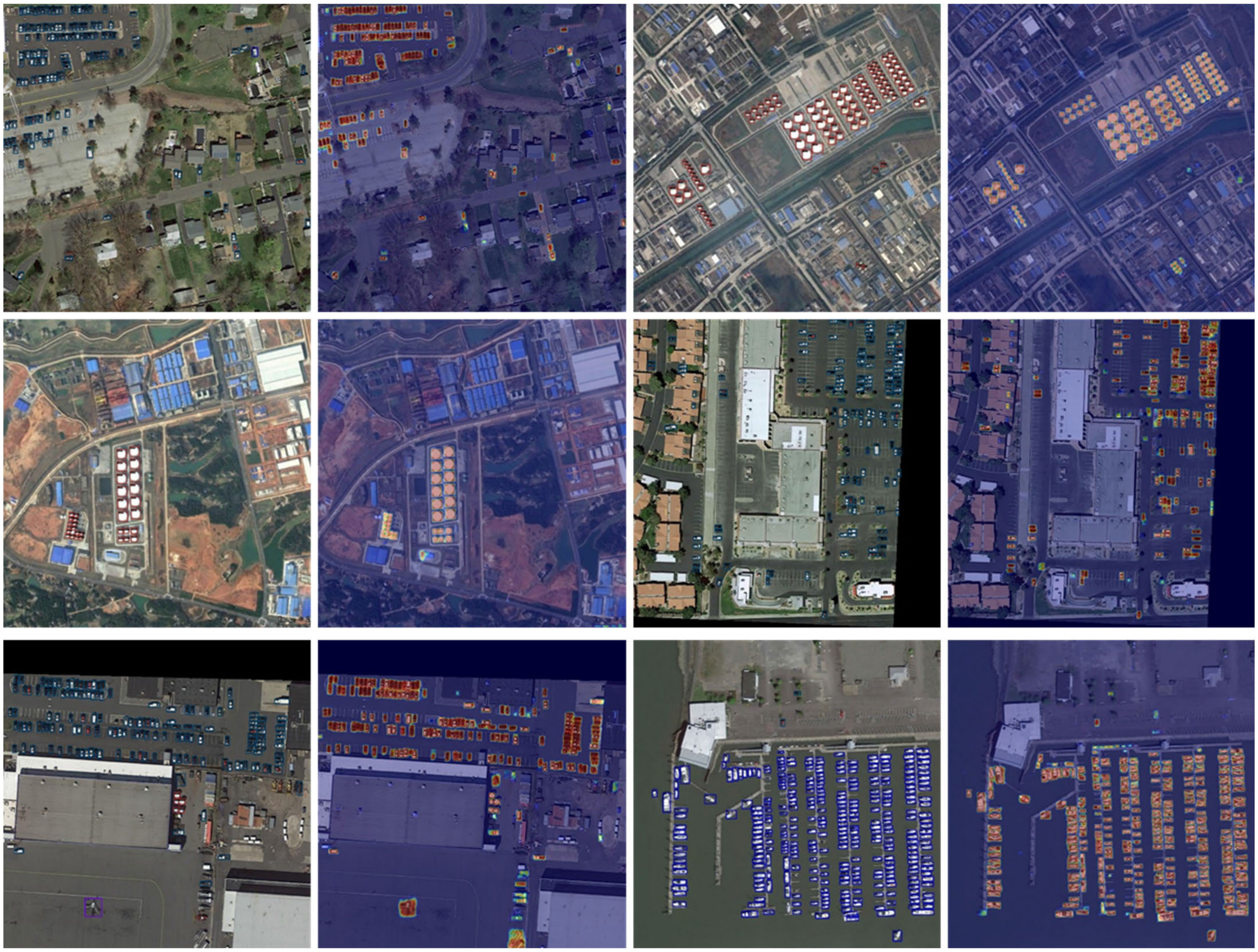


Fig. 3. Visual results of the proposed method. (a) and (c) Detection results of ORFENet, and (b) and (d) regions of interest emphasized by the proposed ORB. As can be seen from the figure, the proposed method can achieve pretty good results for several kinds of objects such as vehicles, storage tanks, and ships.

2) Ablation Study: In this section, we explore the effects of the kernel size of the proposed MRFAFEM, the dynamic weights of different receptive fields in the proposed MRFAFEM, and the impact of λ in the loss function.

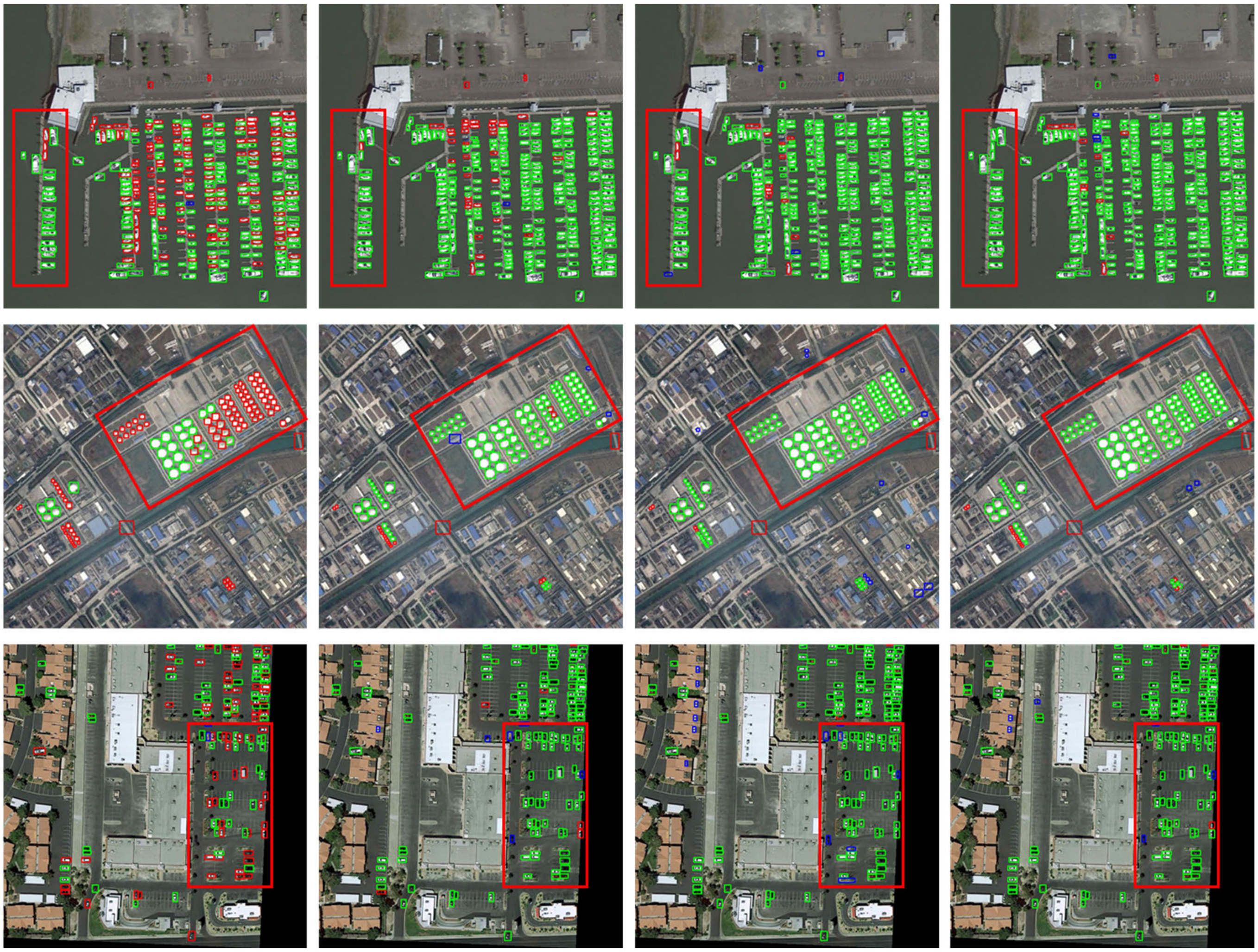
We conducted ablation experiments on the kernel size of depthwise convolution in MRFAFEM. First, we fix the convolution kernel size of the fine-grained branch in MRFAFEM and the convolution kernel of the distant context branch to 21. Then, we modify the convolution kernel size of the close-range context branch to 5, 7, and 9, respectively, to determine the most suitable size for the close-range context branch convolution kernel. In addition, we fix the convolution kernel size of the fine-grained branch and the close-range context branch to 7 and modify the convolution kernel size of the distant context branch to 17, 21, and 25, respectively, to determine the most suitable size for the distant context branch convolution kernel. The experimental results are presented in Table II. From the table, it can be observed that using MRFAFEM can enhance the performance of tiny object detection. When the convolution kernel size of the close-range context branch is 7 and the distant context branch convolution kernel size is 21, the proposed method achieves the best detection performance. Therefore, this article defaults to setting the convolution kernel of the close-range context branch and distant context branch to 7 and 21, respectively.

TABLE VII

PERFORMANCE OF THE PROPOSED ORFENET AND SEVERAL EXCELLENT METHODS ON THE LEVIR-SHIP DATASET. THE PROPOSED ORFENET CAN ACHIEVE THE BEST RESULTS OF 83.8% AP50

Method	AP50(%)	Method	AP50(%)
Efficient-D0 [44]	71.3	Efficient-D2 [44]	80.9
RetinaNet [25]	74.9	FCOS [27]	75.5
SSD [20]	52.9	CenterNet [28]	77.7
Faster R-CNN [34]	70.8	HSFNet	73.6
DRENet [19]	82.4	ORFENet	**83.3**

The impact of the dynamics of multiple receptive field features in the proposed MRFAFEM on the final performance is investigated. The experimental results are shown in Table III. MRF indicates the MRFAFEM without the dynamic weight of multiple receptive fields features, and weight indicates the inclusion of dynamic weight of multiple receptive fields



(a) (b) (c) (d)

Fig. 4. Visual comparisons of the proposed method and other methods. (a) Baseline FCOS. (b) FSANet. (c) Cascade-R-CNN w/NWD-RKA. (d) Proposed ORFENet. The green boxes denote the true-positive predictions, the red boxes denote the false-negative predictions, and the blue boxes denote the false-positive predictions.

features. The table illustrates that the incorporation of the multiple receptive fields features into the baseline enhances the detection performance of tiny objects to some extent. Furthermore, when dynamic weight is applied to the MRF, the ability to detect tiny objects is further improved. This demonstrates the effectiveness of the dynamic weight of different receptive fields in the proposed MRFAFEM.

The weight of different losses has always been a challenging hyperparameter to determine in a multiloss network. Therefore, we investigate the impact of the weight of the ORB in the loss function on the final performance. We conducted several experiments with different weights of 0, 1, 5, 10, and 20. $\lambda = 0$ indicates that the ORB is not included. The experimental results are shown in Table IV. It can be observed that when λ is set to 1 or 5, there is some performance improvement compared with the baseline. When λ is set to 10, the detector achieves a significant performance gain compared with o the baseline. As λ increases, when λ is set to 20, due to the excessive object reconstruction loss, the proportion of the detector loss decreases, resulting in a decline in performance gain. When λ is equal to 10, the detector achieves the best performance. Therefore, we set λ as 10 as the default setting.

3) Comparison With Other Methods: We compare the proposed ORFENet with some excellent methods on the AI-TODv2 dataset. The comparison methods include: DotD [14], Cascade R-CNN w/NWD-RKA [16], RetinNet [25], FCOS [27], RepPoints [30], FoveaBox [31], Faster R-CNN [34], Faster R-CNN w/ATSS [34], [49], Cascade R-CNN [35], DetectoRS [37], TridentNet [45], ATSS [49], and FSANet [61]. The results are shown in Tables V and VI.

From Table IV, it can be observed that the proposed ORFENet can achieve the best results in terms of AP, AP50, AP75, AP_{vt}, AP_t, and AP_s. Specifically, ORFENet obtains an AP of 24.8%, AP50 of 55.4%, AP75 of 18.2%, AP_{vt} of 9.7%, AP_t of 24.4%, and AP_s of 28.7%. Compared with the two-stage DetectoRS, AP increased by 8.7%, AP50 by 19.9%, AP75 by 5.7%, AP_{vt} by 9.6%, AP_t by 11.8%, and AP_s by 0.4%. Compared to the single-stage FSANet, AP increased by 7.2%, AP50 by 10.4%, AP75 by 7.7%, AP_{vt} by 4.3%, AP_t

Fig. 5. Detection results of the proposed ORFENet on the LEVIR-Ship dataset. The proposed method can achieve good performance in different situations.

by 8.6%, and AP_s by 5.8%. From Table VI, it can be observed that the proposed method also achieves the best results in six categories: BR, SH, SP, VE, PE, and WM.

Moreover, to further validate the effectiveness of the proposed method, we have conducted experiments on the LEVIR-Ship dataset and compared it with other state-of-the-art methods, as shown in Table VII. From the table, it is evident that our method achieved the best results, with an AP50 of 83.3%. Compared with the RetinaNet, AP50 improved by 8.4%, and compared with Faster R-CNN, AP50 improved by 12.5%. Moreover, compared with the recent DRENet, AP50 increased by 0.9%. Therefore, this demonstrates the effectiveness of the proposed method.

4) Several Visualization Results: We visualize several results on the AI-TODv2 and LEVIR-Ship datasets.

The visualization results of the proposed ORFENet on the AI-TODv2 dataset are shown in Fig. 3, where all the detection results with confidence greater than 0.3 are displayed. The first and third columns represent the detection results of ORFENet, while the second and fourth columns represent the regions of interest emphasized by the proposed ORB. From the figure, it is evident that our method performs well in detecting and recognizing vehicles, storage tanks, and ships. Moreover, the ORB effectively focuses on the regions of tiny objects and reconstructs them accordingly. Thus, the proposed ORB is effective.

Furthermore, the visualization comparison of different methods is shown in Fig. 4. The compared methods include the best performing two-stage method Cascade-R-CNN w/NWD-RKA, the best performing single-stage method FSANet, and the baseline method FCOS. It is evident from the figure that compared with the baseline FCOS method, the proposed method shows significant improvement. Furthermore, compared with o the other two methods, the proposed method exhibits better results in terms of the number of false alarms and missed detections. Specifically, it can be observed from the red boxes in the first and second rows that only the proposed method has no false alarms or missed detections, while the other two methods both have instances of false alarms and missed detections. In addition, it can be seen from the third row that the proposed method has fewer erroneous detections compared with the other methods. Therefore, the proposed method demonstrates superior detection performance.

The detection results of the proposed ORFENet on the LEVIR-Ship dataset are shown in Fig. 5. As can be seen from the figure, the proposed ORFENet can achieve relatively good performance in different situations, including calm sea and thick cloud. Therefore, the proposed method is effective.

V. Conclusion

In this article, a remote sensing image tiny object detection method is proposed based on object reconstruction and MRFAFEM. The proposed method builds upon the FCOS framework, aiming to alleviate the issue of information loss for tiny objects within deep networks by applying the ORB on the feature map used for detecting tiny objects. In addition, the method uses MRFAFEM to dynamically adjust features from various receptive fields, thereby facilitating the detection of tiny objects. Compared with several state-of-the-art methods, the experimental results demonstrate the effectiveness of the proposed method for the task of tiny object detection in remote sensing image. Although the proposed method performs well in detecting tiny objects, its complexity is relatively high. In the future, reducing the complexity of tiny object detection algorithms is an important research direction.

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